

Problem 8.3.10

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Question: If the eccentricity of an ellipse is $\frac{5}{8}$ and the distance between its foci is 10, then find the latus rectum of the ellipse.

Solution

Given:

- Eccentricity: $e = \frac{5}{8}$
- Distance between foci: $D = 10$

Step 1: Express the foci for standard ellipse

From B.2.7.3, the foci for standard ellipse are at:

$$\mathbf{F} = \pm \sqrt{\frac{f_0}{\lambda_2(1 - e^2)}} \mathbf{e}_1 \quad (2.1)$$

where f_0 is the scalar term in the conic equation and λ_2 is an eigenvalue of matrix $\mathbf{V}$.

Step 2: Distance between foci

The distance between foci is:

$$D = 2 \sqrt{\frac{f_0}{\lambda_2(1 - e^2)}} \quad (2.2)$$

Solution

Step 3: Solve for f_0

From equation (2):

$$\frac{D}{2} = \sqrt{\frac{f_0}{\lambda_2(1 - e^2)}} \quad (2.3)$$

Squaring both sides:

$$\frac{D^2}{4} = \frac{f_0}{\lambda_2(1 - e^2)} \quad (2.4)$$

Therefore:

$$f_0 = \frac{D^2 \lambda_2 (1 - e^2)}{4} \quad (2.5)$$

GStep 4: Eigenvalues for standard ellipse

For the standard ellipse, from the conic section theory:

$$\lambda_1 = \frac{||\mathbf{n}||^2}{1 - e^2} \quad (2.6)$$

$$\lambda_2 = ||\mathbf{n}||^2 \quad (2.7)$$

For the standard form with $||\mathbf{n}||^2 = 1$:

$$\lambda_1 = \frac{1}{1 - e^2} \quad (2.8)$$

$$\lambda_2 = 1 \quad (2.9)$$

Solution

Step 5: Calculate f_0 with $\lambda_2 = 1$

Substituting $\lambda_2 = 1$ into equation (5):

$$f_0 = \frac{D^2(1 - e^2)}{4} \quad (2.10)$$

Therefore:

$$\sqrt{f_0} = \frac{D \sqrt{1 - e^2}}{2} \quad (2.11)$$

Step 6: Matrix formulation for latus rectum The length of latus rectum is given by :

$$\ell = \frac{2 \sqrt{f_0 \lambda_1}}{\lambda_2} \quad (2.12)$$

Solution

Substituting the values of f_0 , λ_1 and λ_2

$$\ell = \frac{D(1 - e^2)}{e} \quad (2.13)$$

Step 7: Substitute the given values

Given $e = \frac{5}{8}$ and $D = 10$:

$$1 - e^2 = 1 - \left(\frac{5}{8}\right)^2 = 1 - \frac{25}{64} = \frac{39}{64} \quad (2.14)$$

Substituting into the latus rectum formula:

$$\ell = \frac{390 \times 8}{64 \times 5} = \frac{3120}{320} = \frac{39}{4} \quad (2.15)$$

Final Answer: The length of latus rectum is :

$$\ell = \frac{39}{4}$$

(2.16)

Graph

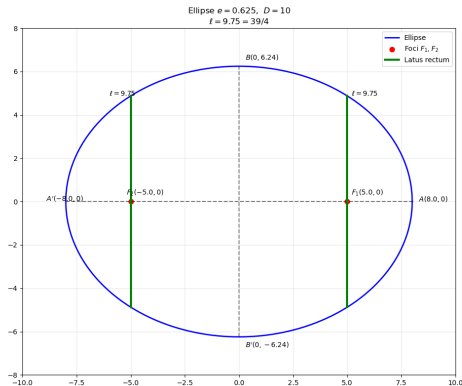


Figure: Vector Representation

```
#include <math.h>
#include "matfun.h"

double semi_major ( double e , double D ){
    return D / ( 2.0 * e );
}

double semi_minor ( double e , double D ){
    double a = semi_major ( e , D );
    return a * sqrt ( 1.0 - e * e );
}

double focal_distance ( double e , double D ){
    return semi_major ( e , D ) * e;
}
```

```
double latus_rectum ( double e , double D ){
    return D * ( 1.0 - e * e ) / e;
}

void ellipse_points(double a , double b ,int N ,double *x ,double
    *y){
    for ( int i = 0 ; i < N ; ++i )
    {
        double t = 2.0 * M_PI * i / N;
        x [ i ] = a * cos ( t );
        y [ i ] = b * sin ( t );
    }
}
```

Python + C Code

```
import ctypes as ct
import numpy as np
import matplotlib.pyplot as plt
from numpy.ctypeslib import ndpointer

lib = ct.CDLL ( './libmatfun.so' )

for f in ( 'semi_major' , 'semi_minor' ,
          'focal_distance' , 'latus_rectum' ):
    getattr ( lib , f ).argtypes = [ ct.c_double , ct.c_double ]
    getattr ( lib , f ).restype = ct.c_double

lib.ellipse_points.argtypes = [
    ct.c_double , ct.c_double , ct.c_int ,
    ndpointer ( ct.c_double , flags = 'C_CONTIGUOUS' ),
    ndpointer ( ct.c_double , flags = 'C_CONTIGUOUS' )
]
```

Python + C Code

```
e , D = 5 / 8 , 10

a = lib.semi_major ( e , D )
b = lib.semi_minor ( e , D )
c = lib.focal_distance ( e , D )
l = lib.latus_rectum ( e , D )

print ( l ) # prints 9.75

N = 1000
x = np.empty ( N )
y = np.empty ( N )

lib.ellipse_points ( a , b , N , x , y )

fig , ax = plt.subplots ( figsize = ( 12 , 8 ) )
```

```
ax.plot ( x , y , 'b-' , lw = 2 )
ax.scatter ( [ c , -c ] , [ 0 , 0 ] , color = 'r' , s = 50 )
ax.plot ( [ c , -c ] , [ 0 , 0 ] , 'r--' , lw = 1 )

ax.plot ( [ c , c ] , [ -1 / 2 , 1 / 2 ] , 'g' , lw = 3 )
ax.plot ( [ -c , -c ] , [ -1 / 2 , 1 / 2 ] , 'g' , lw = 3 )

ax.plot ( [ -a , a ] , [ 0 , 0 ] , 'k--' , alpha = .5 )
ax.plot ( [ 0 , 0 ] , [ -b , b ] , 'k--' , alpha = .5 )
```

```
ax.text ( a + .3 , 0 , f'A ( {a} )' )
ax.text ( -a - 1 , 0 , f"A' ( -{a} )" )
ax.text ( .3 , b + .3 , f'B ( {b:.2f} )' )
ax.text ( .3 , -b - .6 , f"B' ( -{b:.2f} )" )
ax.text ( c + .2 , 1 / 2 , f' = {1:.2f}' )
ax.text ( -c - 1 , 1 / 2 , f' = {1:.2f}' )

ax.set_aspect ( 'equal' )
ax.set_xlim ( -10 , 10 )
ax.set_ylim ( -8 , 8 )
ax.set_title ( f'Ellipse e = {e} D = {D}' )
ax.grid ( alpha = .3 )

plt.tight_layout ( )
plt.show ( )
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

e, D = 5 / 8, 10
l = D * (1 - e ** 2) / e
a = D / (2 * e)
b = a * np.sqrt(1 - e ** 2)
c = a * e

print("The length of latus rectum is: ", l)
t = np.linspace(0, 2 * np.pi, 1000)
x, y = a * np.cos(t), b * np.sin(t)

fig, ax = plt.subplots(figsize=(12, 8))
ax.plot(x, y, 'b-', lw=2, label='Ellipse')
```



```
ax.scatter([c, -c], [0, 0], color='r', s=50, label='Foci $F_1$,  
$F_2$')  
ax.text(c + .2, 0.3, f'$F_1({c}, 0)$')  
ax.text(-c - .2, 0.3, f'$F_2({-c}, 0)$')  
  
ax.plot([-a, a], [0, 0], 'k--', alpha=.5)  
ax.plot([0, 0], [-b, b], 'k--', alpha=.5)  
  
ax.text(a + .3, 0, f'$A({a}, 0)$')  
ax.text(-a - .3, 0, f'$A\ '(-{a}, 0)$')  
ax.text(0.3, b + .3, f'$B(0, {b:.2f})$')  
ax.text(0.3, -b - .3, f'$B\ '(0, -{b:.2f})$')
```

```
ax.plot([c, c], [-1 / 2, 1 / 2], 'g', lw=3, label='Latus rectum')
ax.plot([-c, -c], [-1 / 2, 1 / 2], 'g', lw=3)
ax.text(c + .2, 1 / 2, f'$\\ell = {1:.2f}$')
ax.text(-c - 1.0, 1 / 2, f'$\\ell = {1:.2f}$')

ax.set_aspect('equal')
ax.set_xlim(-10, 10)
ax.set_ylim(-8, 8)
ax.set_title(f'Ellipse $e={e}$, $D={D}$\\n$\\ell = {l} = 39/4$')
ax.grid(alpha=.3)
ax.legend()
plt.tight_layout()
plt.savefig("fig1.png")
plt.show()
```